

Pivot: Evaluating Contextual Reasoning Under Lexical Ambiguity

Proposal

I propose a benchmark for testing whether models can preserve, resolve and appropriately reweight genuinely ambiguous meanings as contextual evidence changes.

Most evals assume either a single correct answer or a clearly specified task objective. Many real-world interactions are harder: a word or phrase may support several plausible meanings, and the appropriate interpretation depends on evidence distributed across the immediate text, prior conversation, cultural context or later information. A reliable model should neither collapse ambiguity too early nor preserve unnecessary possibilities once enough evidence exists to make a judgment.

The method draws on the poetic concept of *kakekotoba* — a pivot word that supports multiple meanings — as a controlled way to test how models interpret and reweight ambiguity.

Pivot would use a held-out corpus of original short-form poetry containing deliberately polysemous words, phrases and formal hinges. Poetry provides a useful controlled environment because ambiguity can be manipulated at very small scale through lexical choice, line breaks, punctuation, typography, titles and contextual framing.

For example:

**Stationary
leaves echoing
the last train**

The ambiguity enables evidence-based reasoning. Leaves pivots lexically (foliage, pages, departing); stationary pivots phonetically (stationery, with station surfacing against train); the line pivots syntactically, since leaves can also be read as a verb: as something still, departing, echoing. Replacing echoing with echoes strengthens the verb reading while leaving the other pivots untouched. A model's weighting should shift accordingly.

The benchmark would test models across multiple turns:

1. identify the smallest defensible set of plausible interpretations;
2. anchor each interpretation in textual evidence;
3. weight the available readings;
4. incorporate relevant cultural or conversational context;
5. respond to counter-evidence or an alternative reading;
6. revise weighting and uncertainty;
7. resist irrelevant or misleading context.

To reduce contamination, the body of work used would consist of previously unpublished human-authored and machine-generated items created specifically for evaluation.

Performance would be evaluated on interpretive coverage, evidence anchoring, context discrimination, calibration, revision proportionality, form sensitivity and cross-turn consistency.

The benchmark would focus not simply on whether a model changes its answer, but whether the direction and magnitude of revision track the evidential value of new information.

The initial goal would be to determine whether controlled poetic ambiguity provides a reliable and discriminating testbed for **context-sensitive reasoning under uncertainty**.

If successful, the methodology could transfer to professional domains where models must interpret shorthand, conflicting cues, culturally specific language and information distributed across prior interactions.